

Lab 7 Report: Usability Evaluation

MMAC Study Hub and Aura Screen

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1 Introduction

We evaluated two products from the Multimodal Affective Computing (MMAC) study. The **Study Hub** is a website that introduces the study and showcases its thermal and ECG data in interactive dashboards. **Aura Screen** is a patient health-screening app for phones and screening kiosks: sign up, scan at a station, read the report.

Three problems drove this evaluation. Aura Screen was built for a desktop browser, so its phone/kiosk view could only be shown through browser developer tools, and screens needed scrolling. The hub linked to separate Streamlit sensor dashboards that stopped launching after a while because of deprecated dependencies. And the project’s goal moved from *recording* data to *showcasing* it, which the recording dashboards did not fit.

2 Test plan

Tasks.

- **T1 (Aura Screen):** create an account, finish onboarding, start a scan and read your heart rate on the report.
- **T2 (Study Hub):** find the ECG data, compare heart rate at rest vs city driving, and start a data-access request.

T1 is the app’s core end-to-end journey; T2 covers the hub’s two jobs, showing data and controlling its release, and requires reading a chart.

Users. Five people from outside the team (P1–P5), tested on 28–29 September 2026. All five were technical/research students, the group closest to the hub’s researcher audience; only P4 had used a similar health or data app before.

Procedure. About 12 minutes each: consent, a think-aloud warm-up, T1, T2, the SUS questionnaire and two debrief questions. One of us facilitated; the other observed and filled in the test form. All sessions ran on a laptop; Aura Screen was used in its phone-sized view in the browser, not on a real phone.

Metrics. Completion with or without help (effectiveness), think-aloud confusion points (why), and SUS (standard satisfaction score; 68 is average). The form also had time and error fields, but these were not filled in during the sessions, so no timing or error counts are reported.

3 Findings

3.1 User testing

	T1 Aura Screen	T2 Study Hub	SUS
P1	Help	Help	77.5
P2	Pass	Pass	62.5
P3	Help	Help	67.5
P4	Pass	Pass	90.0
P5	Pass	Help	67.5
Completed	5/5	5/5	
Unaided	3/5	2/5	73.0

Pass = completed alone; Help = completed with help.

Every participant finished both tasks; nobody failed. Half of the attempts needed a hint, most of them on the website (3 of 5). The mean SUS of **73.0** (range 62.5–90) is above the 68 average. SUS rated both products together. The lowest item was confidence (Q9, mean 3.6/5), matching the hints needed.

Aura Screen (T1). P2, P4 and P5 called it easy (“self-explanatory”; P4 liked the step bar). P1’s browser blocked the camera: P1 was stuck until shown the *Continue without camera* button (U1). P3 missed the consent tick box in the long record form and had to go back (U2). P5 paused to read the scan instructions.

Study Hub (T2). The resting heart rate was found quickly (P1: 79 bpm on the top card). The city-driving rate was not shown as a number, so P1 and P3 had to read it off the chart (U3). P3 found the ECG page only through the Modalities menu, and P5 did not see that the menu was clickable (U4). P2 did not know what ECG meant at first (U5). P1 found scrolling past the large charts slow (U6). On the access request, P4 was unsure whether the supervisor fields applied to them (U7), and P5 found the agreement text long and dense (U8). P2 did not realise that apps such as Aura Screen are built on the data (U9). P4 found the three-step form and draggable widgets clear.

3.2 Heuristic evaluation

Method. Each of us walked through T1 and T2 on our own and noted every problem against Nielsen’s ten heuristics. For each problem we recorded where it occurs (screen and task step), the heuristic it violates and a severity: 0 not a problem, 1 cosmetic, 2 minor, 3 major, 4 catastrophe. We then merged the two lists and removed duplicates. The inspection covered the builds from before this lab’s changes. Everything was checked on a laptop, so A4–A6, which only affect touch screens, were judged from the design rather than observed.

Results. The merged list has 17 problems (Table 1): one catastrophe (W1), six major, seven minor and three

Table 1: Heuristic evaluation: merged findings from both evaluators (inspection).

ID	Screen / step	Problem	Heuristic violated	Sev.
Aura Screen				
A1	Any screen, desktop browser	Phone/kiosk view only reachable through browser developer tools.	7. Flexibility & efficiency; 4. Consistency & standards	3
A2	Scan Hub (T1, start scan)	Core screens scroll in the phone view: Scan Hub is +454 px taller than a 390×844 screen.	8. Aesthetic & minimalist design	3
A3	Session report (T1, read HR)	Third heart-rate tile cut off at phone width.	8. Aesthetic & minimalist design	3
A4	Navigation, all screens	Navigation sits at the top, out of thumb reach, unlike phone-app convention.	4. Consistency & standards	2
A5	Sign-up, record setup (T1)	13 px inputs make iOS zoom in on every tap.	4. Consistency & standards	2
A7	Kiosk layout, all screens	Kiosk screens leave large empty areas.	8. Aesthetic & minimalist design	2
A6	Buttons and cards (touch)	Hover styles “stick” after a tap, so a tapped item looks still selected.	1. Visibility of system status	1
Study Hub website				
W1	Home → sensor dashboard link (T2, find data)	Links lead nowhere when a dashboard server is down or fails to launch (deprecated versions); no message explains why.	9. Recover from errors; 1. Visibility of system status	4
W2	Sensor dashboards (T2, compare HR)	Dashboards are recording tools; visitors only want to view data.	2. Match with the real world	3
W3	Sensor dashboards (T2)	Separately styled Streamlit apps look and work differently from the site.	4. Consistency & standards	3
W4	Opening the site	Site fails to start (stray build cache); no run instructions.	10. Help & documentation	3
W5	Home page sections	Placeholder “lorem ipsum” text.	8. Aesthetic & minimalist design	2
W6	Data download (T2, request data)	Data offered as open downloads with no terms of use.	5. Error prevention	2
W7	ECG charts (T2, compare HR)	Charts work with a mouse only.	7. Flexibility & efficiency	2
W9	Request access, submit (T2)	Sending relies on an email app being set up; nothing happens without one.	9. Recover from errors	2
W8	ECG dashboard widgets	Widgets can be dragged, but nothing shows this.	6. Recognition rather than recall	1
W10	Thermal page	Sample temperatures are illustrative, not measured.	2. Match with the real world	1

cosmetic. The most-violated heuristics were 8 (minimalist design) and 4 (consistency), with four problems each. Most came from Aura Screen being a desktop layout squeezed into a phone-sized screen. No violations of 3 (user control and freedom) were found.

Inspection vs user testing. The two methods found different problems. Inspection caught layout, platform and robustness problems that users seldom mention, such as dead links, iOS zoom and sticking hover styles. Testing found nine further problems (Table 2). These were mostly about understanding: the missing per-phase numbers, jargon, the dense agreement and the blocked camera. Only one problem was close to an inspection finding: charts being hard to use (W7) and reading values off a chart (U3).

3.3 Accessibility check

A quick manual check of both products, looking for what a user would notice. We read each screen on a laptop (Aura Screen in its phone-sized view), used the website

with the keyboard only, and checked that images and icon buttons are labelled. No tools or measurements.

Table 2: Problems found only in user testing (Section 3.1), rated on the same scale.

ID	Screen / step	Problem (participants)	Heuristic violated	Sev.
U1	Scan step (T1, start scan)	Camera blocked; the way past it (<i>Continue without camera</i>) is easy to miss, so the scan cannot start (P1).	9. Recover from errors; 1. Visibility of system status	3
U3	ECG dashboard (T2, compare HR)	Per-phase heart rate (rest vs city) not shown as a number; users read it off the chart (P1, P3).	6. Recognition rather than recall	3
U2	Record setup (T1)	Long form; the consent box was skipped and only caught on submit (P3).	5. Error prevention	2
U4	Home → Modalities menu (T2, find data)	Modality pages hard to find; the menu does not look clickable (P3, P5).	6. Recognition rather than recall	2
U5	ECG dashboard (T2)	Jargon (“ECG”) and charts with no caption saying what they show (P2, P3).	2. Match with the real world; 10. Help & documentation	2
U6	ECG dashboard, scrolling	Scrolling lags past the large charts (P1).	7. Flexibility & efficiency	2
U7	Request access, details step (T2)	Unclear which fields (supervisor) apply to whom (P4).	5. Error prevention	2
U8	Request access, agreement step (T2)	Agreement is a wall of legal text with no summary (P5).	8. Aesthetic & minimalist design	2
U9	Home page	Apps built on the data are easy to miss; the hub reads as “research only” (P2).	8. Aesthetic & minimalist design	1

Check	What we saw		Issues	Planned change (in priority order)
Text, website	Main text is clear, but faded grey text (inactive menu items, notes on cards) is hard to read.	Fail	U3	Print mean heart rate per phase (rest, city) beside the chart.
Text, charts	Axis labels are small and pale; hard to read at a glance.	Fail	U1	Explain a blocked camera up front; <i>Continue without camera</i> as the main button.
Text, Aura	Body text and buttons are clear; small captions and pink accent text are faint.	Part.	Readability	Darken faded grey, caption and chart-label text in both products.
Text size	Some labels and badges are very small, especially on charts.	Part.	U4, U5	Modality cards on the home page; a one-line caption under each chart; a plain-language line on what ECG/thermal measure.
Keyboard	Tab reaches every widget and Alt+arrows moves it; the request dialog keeps focus and Esc closes it; charts cannot be explored without a mouse.	Part.	U7, U8	Mark optional fields; supervisor fields for students only; bullet summary above the agreement.
Labels	Images have descriptions, but the theme toggle is an unlabelled icon, so a screen reader cannot name it.	Part.	U2	Flag missed required fields (consent) in place.
			U6	Render off-screen charts lazily.
			W7, kbd	Keyboard access and a data table per chart.
			U9, labels	Link “Built with the data” from the hero; name the theme toggle; enlarge the smallest text.

Limitations. All five participants were technical students, so non-technical (patient-like) users still need testing. Everything ran on a laptop, so Aura Screen still needs testing on real phones and kiosks. Task times and errors were not recorded. We will address all three in the next round, when we re-test the Planned changes.

3.4 Live demo

Aura Screen on a laptop: the old build needs dev-tools emulation and Scan Hub scrolls; the phone/kiosk frame fits the same screen without scrolling (A1, A2).

4 Improvement plan

Ranked by severity, then by how many participants hit the issue. “Ready” = prototype built and used in these sessions; “Planned” = next iteration.

Ready: W1–W3 showcase pages replace the Streamlit links; A1–A5, A7 phone/kiosk-first Aura Screen (device frame, no-scroll screens, bottom tabs, 16px inputs, fixed report); W4 one-command launcher; W6 access request with a data-use agreement.